Tutorial 2

DFT and Discrete Fourier Transforms

SC3102: Signals, Systems, and Transforms

Question 1

Roots of Unity and the DFT Matrix

Recall: Roots of Unity and Conjugation

- Recall that the roots of unity are the roots of:

$$z^N = 1$$

- The **primitive** root of unity is defined as:

$$\omega = e^{\mathrm{j}2\pi/N}$$

- The primitive root generates all other roots of unity as:

$$z_k = \omega^k = e^{\mathrm{j}2\pi k/N}$$

Recall: DFT Analysis and Synthesis

- The roots of unity defined the DFT analysis and synthesis equations
- For $\boldsymbol{x} \in \mathbb{C}^N$, let $G_{n,k} = \omega^{nk}$. DFT analysis gives

$$\boldsymbol{X} = G^{\mathrm{H}}\boldsymbol{x}, \qquad X[k] = \sum_{n=0}^{N-1} x[n]\omega^{-kn}.$$

- Its inverse recovers the entries of the vector:

$$\boldsymbol{x} = \frac{1}{N} G\boldsymbol{X}, \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\omega^{kn}.$$

Question 1(a): Primitive Root and DFT Matrices

- Fix $N = 4$.
- Compute the primitive root of unity:

$$z^N = 1.$$

- Write out the DFT analysis and synthesis matrices.

1(a): Primitive Root and Analysis Matrix

- For $N = 4$, the primitive root is

$$\boxed{\omega = \mathrm{e}^{\mathrm{j}\pi/2} = \mathrm{j}}, \qquad \omega^{-1} = -\mathrm{j}.$$

- The analysis matrix is G^{H}, with $(G^{\mathrm{H}})_{k,n} = \omega^{-kn}$:

$$\boldsymbol{X} = G^{\mathrm{H}}\boldsymbol{x}, \qquad \boxed{G^{\mathrm{H}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix}}.$$

1(a): The Synthesis Matrix

- The synthesis matrix is $G/4$, with entries $G_{n,k} = \omega^{kn}$:

$$\boldsymbol{x} = \frac{1}{4} G \boldsymbol{X}, \qquad \boxed{\frac{1}{4} G = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \end{bmatrix}.}$$

- From the lecture, $G^{-1} = G^{\mathrm{H}}/N$

Question 1(b): Compare the Rows of the Matrices

- Consider the analysis and synthesis matrices
- What do you observe about rows 1 and 3 of the two matrices?

1(b): Compare Rows 1 and 3

- In the analysis matrix:

$$(G^{\mathrm{H}})_{1,:} = [1, -\mathrm{j}, -1, \mathrm{j}], \qquad (G^{\mathrm{H}})_{3,:} = [1, \mathrm{j}, -1, -\mathrm{j}].$$

- These two rows are conjugates as seen in the lecture
- We know that g_k and g_{N-k} are conjugates
- The same is true for the synthesis equation

Question 1(c): DFT of a Real Input Vector

- Let $\boldsymbol{x} \in \mathbb{R}^4$ and let $\boldsymbol{X}$ be its DFT.
- What can you conclude about $\boldsymbol{X}$?
- In particular, comment on the coefficients $X[1]$ and $X[3]$.

1(c): The DFT of a Real Input Vector

- Suppose $\boldsymbol{x}$ is real and $\boldsymbol{X} = G^{\mathrm{H}}\boldsymbol{x}$

$$X[1] = x[0] - \mathrm{j}x[1] - x[2] + \mathrm{j}x[3],$$
$$X[3] = x[0] + \mathrm{j}x[1] - x[2] - \mathrm{j}x[3].$$

- Therefore $\boxed{X[3] = X[1]^*}$.
- In generaly, $X[k]$ and $X[N-k]$ are conjugates for real $\boldsymbol{x}$

1(d): Recover the Missing Data

- A **real** vector and its DFT satisfy

$$\boldsymbol{v} = \begin{bmatrix} v[0] \\ v[1] \\ 0 \\ v[3] \end{bmatrix}, \qquad \boldsymbol{V} = \begin{bmatrix} 4 \\ 3+3\mathrm{j} \\ A \\ B \end{bmatrix}.$$

- Determine the missing coefficients A, B and $\boldsymbol{v}$.

1(d): Recover the Missing Data

- Since $\boldsymbol{v}$ is real,

$$\boxed{B = V[3] = V[1]^* = 3 - 3\mathrm{j}}, \qquad A \in \mathbb{R}.$$

- To recover A recall row 2 of the synthesis matrix $G/4$:

$$\frac{1}{4}\begin{bmatrix}1 & -1 & 1 & -1\end{bmatrix} \implies v[2] = \frac{1}{4}(V[0] - V[1] + V[2] - V[3])$$

- Use the known sample $v[2] = 0$ in the synthesis equation:

$$0 = 4v[2] = 4 - (3 + 3\mathrm{j}) + A - (3 - 3\mathrm{j}) = A - 2.$$

1(d): Recovering the Original Vector

- We have the DFT $\boldsymbol{V}$
- We can now recover $\boldsymbol{v}$:

$$\boldsymbol{v} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \end{bmatrix}\begin{bmatrix} 4 \\ 3+3\mathrm{j} \\ 2 \\ 3-3\mathrm{j} \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 2 \end{bmatrix}$$

Question 2

DTFS and Periodic Reconstruction

Recall: Periodic Signals

- A signal $x[n]$ is periodic if it satisfies:

$$x[n+N] = x[n]$$

  for some integer N
- The smallest such N is the fundamental period
- The DTFS for a periodic x (with period N) is:

$$X[k] = \sum_{n=0}^{N-1} x[n] \mathrm{e}^{-\mathrm{j}2\pi kn/N}, \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \mathrm{e}^{\mathrm{j}2\pi kn/N}$$

- Synthesis equation matches the input x on $0, ..., N-1$

Question 2: DTFS of Sinusoids

- Consider the two signals:

$$a[n] = \cos\left(\frac{2\pi n}{8}\right), \qquad b[n] = \sin\left(\frac{2\pi n}{6} + \frac{\pi}{3}\right).$$

- Find their fundamental periods.
- Then find their DTFS coefficients **without the analysis sum**.

2(a): Fundamental Periods

- For $a[n]$:

$$\frac{2\pi N}{8} = 2\pi m \quad \Longrightarrow \quad \boxed{N_a = 8}.$$

- For $b[n]$:

$$\frac{2\pi N}{6} = 2\pi m \quad \Longrightarrow \quad \boxed{N_b = 6}.$$

Question 2(b): Find the DTFS Transforms

- Recall the real discrete-time signals

$$a[n] = \cos\left(\frac{2\pi n}{8}\right), \qquad b[n] = \sin\left(\frac{2\pi n}{6} + \frac{\pi}{3}\right).$$

- Compute their DTFS coefficients **without using the analysis sum**.
- Use the fundamental periods found in part (a).

2(b): The DTFS of a

- We will expand a as the sum of complex exponentials:

$$a[n] = \cos\left(\frac{2\pi n}{8}\right) = \frac{e^{\mathrm{j}2\pi n/8} + e^{-\mathrm{j}2\pi n/8}}{2} = \frac{1}{2}\mathrm{e}^{\mathrm{j}2\pi n/8} + \frac{1}{2}\mathrm{e}^{\mathrm{j}2\pi(7)n/8}.$$

- Let's compare to the synthesis equation:

$$a[n] = \frac{1}{8}\sum_{k=0}^{7} A[k]\mathrm{e}^{\mathrm{j}2\pi kn/8} \implies \boxed{A[1] = 4, \qquad A[7] = 4.}$$

- All other coefficients are zero.

2(b): The DTFS of b

- We can similarly expand the sin signal b:

$$b[n] = \frac{e^{j\pi/3}}{2j} e^{j2\pi n/6} - \frac{e^{-j\pi/3}}{2j} e^{-j2\pi n/6} = \frac{e^{j\pi/3}}{2j} e^{j2\pi n/6} - \frac{e^{-j\pi/3}}{2j} e^{j2\pi(5)n/6}.$$

- Let's compare to the synthesis equation as before:

$$b[n] = \frac{1}{6} \sum_{k=0}^{5} B[k] e^{j2\pi kn/6} \implies \boxed{B[1] = 3e^{-j\pi/6}, \qquad B[5] = 3e^{j\pi/6}}.$$

- All other coefficients are zero.

Question 2(c): Synthesis with $N = 12$

- Consider the same real discrete-time signals

$$a[n] = \cos\left(\frac{2\pi n}{8}\right), \qquad b[n] = \sin\left(\frac{2\pi n}{6} + \frac{\pi}{3}\right).$$

- Fourier coefficients ($N = 12$) are computed using indices $0, \ldots, 11$.
- Does the corresponding synthesis recover the original signals?

2(c): The Synthesis Equation

- Inverse synthesis recovers the input signal at $n = 0, \ldots, 11$
- Outside these indices, it **periodically** repeats this block
- When does this periodic repetition match the original signal?

When N is a period of the singal!

2(c): Periods of a and b

- **For $a[n]$: No!** Twelve is not a period:

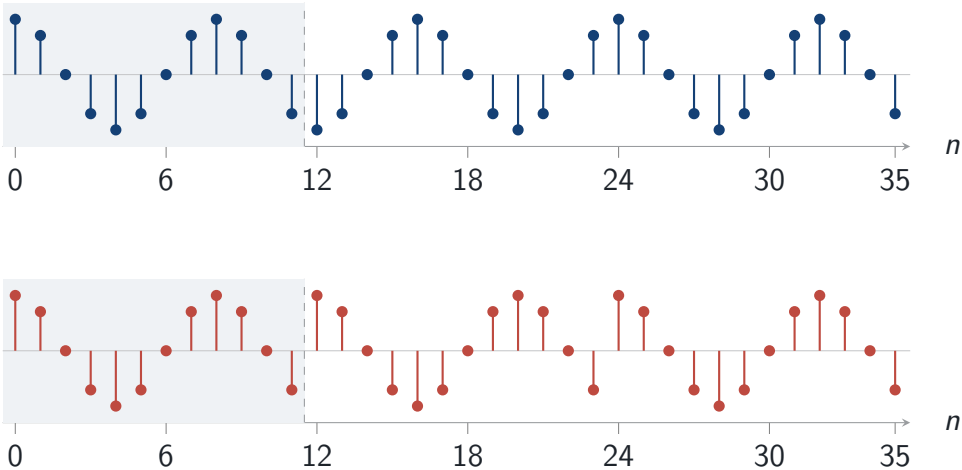
$$a[0] = 1, \qquad a[12] = \cos(3\pi) = -1.$$

- **For $b[n]$: Yes!** Twelve is a period:

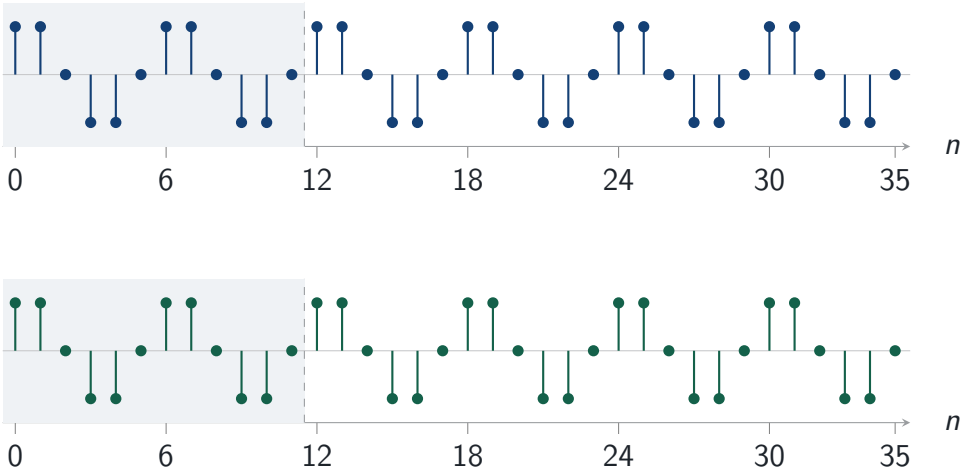
$$b[n+12] = \sin\left(\frac{2\pi n}{6} + 4\pi + \frac{\pi}{3}\right) = b[n].$$

- In fact, 12 is *twice* the fundamental period for b

2(c): Comparing $a[n]$ with Its Synthesis



2(c): Comparing $b[n]$ with Its Synthesis



Question 3

DTFT and Approximation with Finite Samples

Recall: DTFT

- For a discrete-time signal $x[n]$, its DTFT is

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] \mathrm{e}^{-\mathrm{j}\Omega n}.$$

- Frequency Ω is a continuous variable in $[0, 2\pi)$
- $X(\Omega)$ is 2π-periodic in Ω: $X(\Omega + 2\pi) = X(\Omega)$
- For the DTFT to be defined, we used **absolute summability**:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$

Question 3: DTFT of Aperiodic Signals

- Let $a \in \mathbb{R} \setminus \{0\}$. Consider the real discrete-time signals

$$x[n] = a^{-n}u[n], \qquad y[n] = (n+1)a^{-n}u[n].$$

- For what values of a are the DTFT of x and y defined?

3(a): When Are the DTFTs Defined?

- Let $r = |a|^{-1}$. Let's take the first signal:

$$x[n] = a^{-n}u[n] \implies \sum_{n=-\infty}^{\infty} |x[n]| = \sum_{n=0}^{\infty} r^n$$

- The above sum is finite when:

$$|r| < 1 \implies |a| > 1$$

by the geometric sum formula

3(a): When Are the DTFTs Defined?

- Now, let's take the second signal:

$$y[n] = (n+1)a^{-n}u[n]$$

- Instead, let's consider $x * x$:

$$(x * x)[n] = \sum_{m=-\infty}^{\infty} x[m]x[n-m] = \sum_{m=-\infty}^{\infty} a^{-m}u[m]a^{m-n}u[n-m]$$
$$= a^{-n}\sum_{m=-\infty}^{\infty} u[m]u[n-m] = a^{-n}(n+1)u[n] = y[n]$$

as $u[m]u[n-m] = 1$ when $0 \leq m \leq n$.

3(a): When Are the DTFTs Defined?

- We can now use this relationship and the triangle inequality:

$$\sum_{m=-\infty}^{\infty} |y[m]| = \sum_{m=-\infty}^{\infty} \left| \sum_{\ell=-\infty}^{\infty} x[\ell]x[m-\ell] \right| \leq \sum_{m=-\infty}^{\infty} \sum_{\ell=-\infty}^{\infty} |x[\ell]|\,|x[m-\ell]|$$

$$= \sum_{\ell=-\infty}^{\infty} |x[\ell]| \sum_{m=-\infty}^{\infty} |x[m-\ell]| = \sum_{\ell=-\infty}^{\infty} |x[\ell]| \sum_{m=-\infty}^{\infty} |x[m]|$$

- Therefore:

$$\sum_{m=-\infty}^{\infty} |y[m]| \leq \left(\sum_{m=-\infty}^{\infty} |x[m]| \right)^2$$

3(a): When Are the DTFTs Defined?

- From before,

$$\sum_{m=-\infty}^{\infty} |y[m]| \leq \left(\sum_{m=-\infty}^{\infty} |x[m]| \right)^2$$

- Now observe that:

$$\sum_{m=-\infty}^{\infty} |x[m]| < \infty \implies \sum_{m=-\infty}^{\infty} |y[m]| < \infty$$

- Hence, we get the same criterion!

$$|a| > 1$$

Question 3(b): Compute the DTFTs

- Recall the real discrete-time signals, with $a \neq 0$:

$$x[n] = a^{-n}u[n], \qquad y[n] = (n+1)a^{-n}u[n].$$

- Compute the DTFTs of x and y.
- How are they related?

3(b): The DTFT of x

- Assume $|a| > 1$ and let $q = a^{-1}\mathrm{e}^{-\mathrm{j}\Omega}$, so $|q| < 1$.

$$\begin{aligned} X(\Omega) &= \sum_{n=0}^{\infty} a^{-n}\mathrm{e}^{-\mathrm{j}\Omega n} \\ &= \sum_{n=0}^{\infty} q^n = \boxed{\frac{1}{1 - a^{-1}\mathrm{e}^{-\mathrm{j}\Omega}}}. \end{aligned}$$

- The expression is continuous and 2π-periodic function of Ω.

3(b): The DTFT of y

- Remember that $y = x * x$
- Therefore, we get by the convolutional property:

$$Y(\Omega) = X(\Omega)^2 = \boxed{\left(\frac{1}{1 - a^{-1}\mathrm{e}^{-\mathrm{j}\Omega}}\right)^2}.$$

3(c): Approximation with Finite Samples

- Collecting infinitely many samples is expensive
- We usually compute DTFTs with finitely many samples:

$$X_N(\Omega) = \sum_{n=0}^{N} x[n]\mathrm{e}^{-\mathrm{j}\Omega n}.$$

- For $a = 2$, how many samples make the error at most 0.01?
- What happens when a approaches 1?

3(c): Analyzing the Error

- Let's look at the error between X_N and X:

$$|X(\Omega) - X_N(\Omega)| = |\sum_{m=N+1}^{\infty} a^{-m} e^{-\mathrm{j}\Omega m}| \leq \sum_{m=N+1}^{\infty} |a^{-m} e^{-\mathrm{j}\Omega m}|$$
$$= \sum_{m=N+1}^{\infty} |a^{-m}| = \frac{|a|^{-N}}{|a| - 1}$$

- For $a = 2$, we need to set N so that:

$$2^{-N} < 0.01.$$

3(c): Determining N

- We require $2^{-N} \leq 0.01$ or $N \geq \log_2(100)$.

$$N = 6: \quad \frac{1}{64} = 0.015625 > 0.01,$$

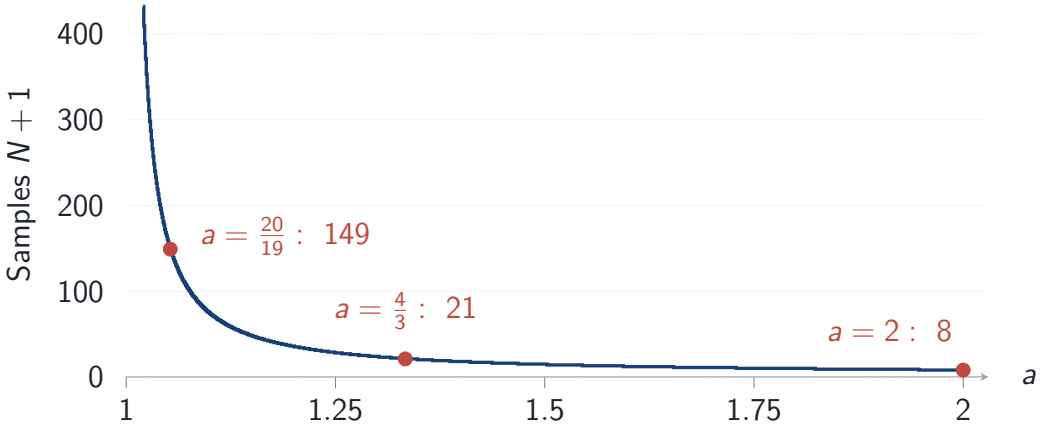
$$N = 7: \quad \frac{1}{128} = 0.0078125 < 0.01.$$

- Hence we need $N_{\text{min}} = 7$ (or 8 samples)

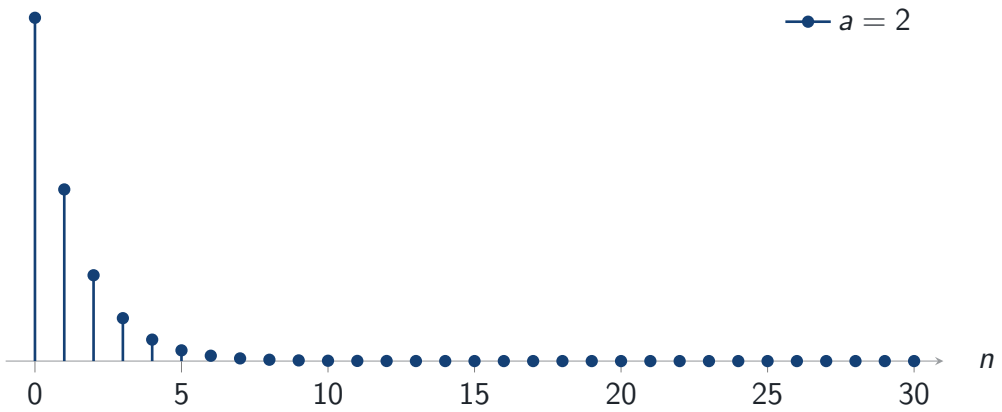
3(c): What happens when a approaches 1?

- Remember that:
$$\max_{\Omega} |X - X_N| \leq \frac{|a|^{-N}}{|a| - 1}.$$
- To make this at most 0.01, choose an integer N satisfying
$$N \geq \frac{\log(100/(|a| - 1))}{\log |a|}.$$
- As $a \downarrow 1$, we need larger and larger N!

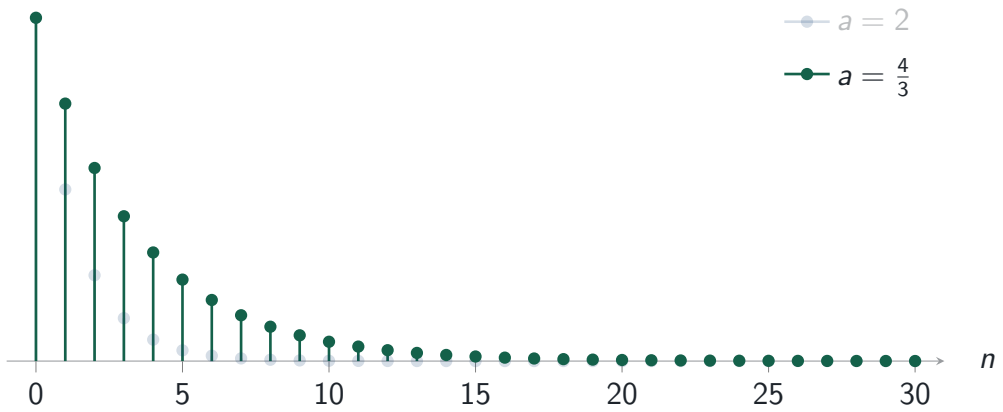
3(c): Samples Needed for Error ≤ 0.01



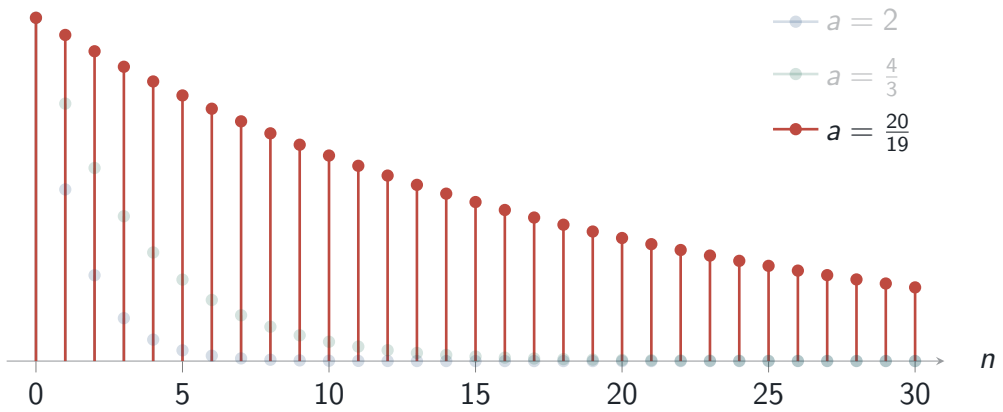
3(c): $x[n] = a^{-n}u[n]$ as $a \downarrow 1$



3(c): $x[n] = a^{-n}u[n]$ as $a \downarrow 1$



3(c): $x[n] = a^{-n}u[n]$ as $a \downarrow 1$



Question 4

Linear and Circular Convolution

Recall: Linear Convolution

- Recall linear convolution was defined as follows:

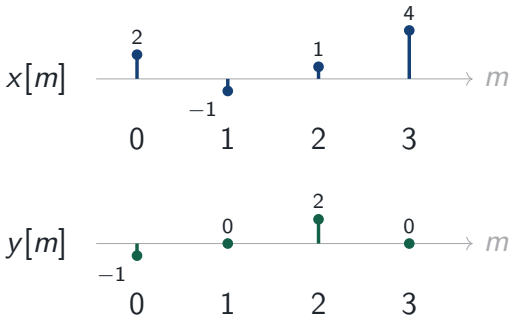
$$(x * y)[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m]$$

- We also interpreted this by transforming the second signal y
- We started by reversing and shifting y
- Multiplied the resultant signal with x and summed over the products

Question 4: Convolving Two Signals

- We are given the following signals beginning at $n = 0$:

$$x = [2, -1, 1, 4], \qquad y = [-1, 0, 2].$$



Question 4(a): Linear Convolution

- The real discrete-time signals are

$$x = [2, -1, 1, 4], \qquad y = [-1, 0, 2].$$

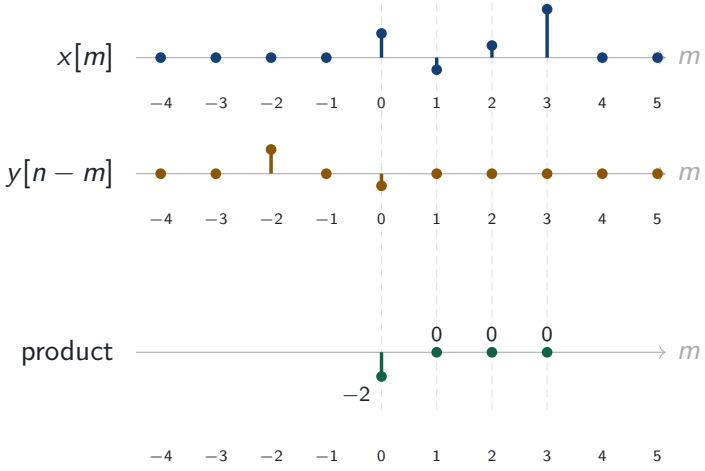
- Compute the **linear convolution** of x and y:

$$(x * y)[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m]$$

- For $n < 0$ and $n \geq 6$, the convolution is 0

4(a): Computing the Linear Convolution

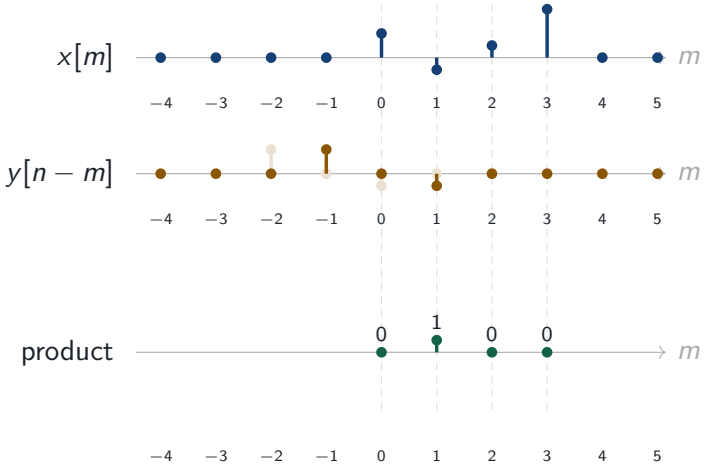
- Keep $x[m]$ fixed and reverse, and shift y



$n = 0$
-2

4(a): Computing the Linear Convolution

- Keep $x[m]$ fixed and reverse, and shift y

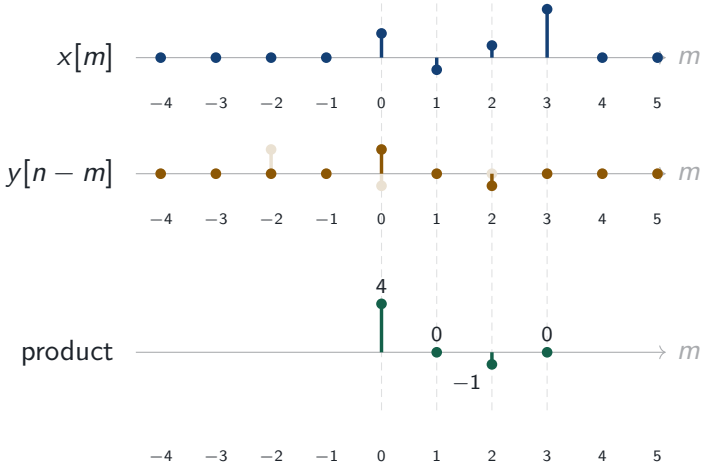


$n = 1$

1

4(a): Computing the Linear Convolution

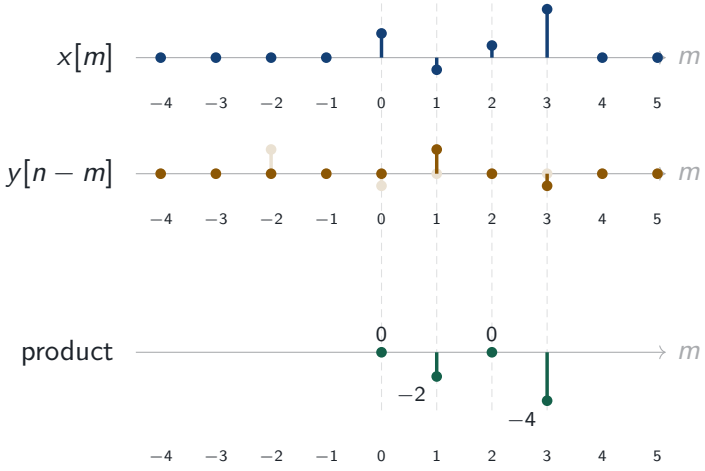
- Keep $x[m]$ fixed and reverse, and shift y



$$n = 2$$
$$4 - 1 = 3$$

4(a): Computing the Linear Convolution

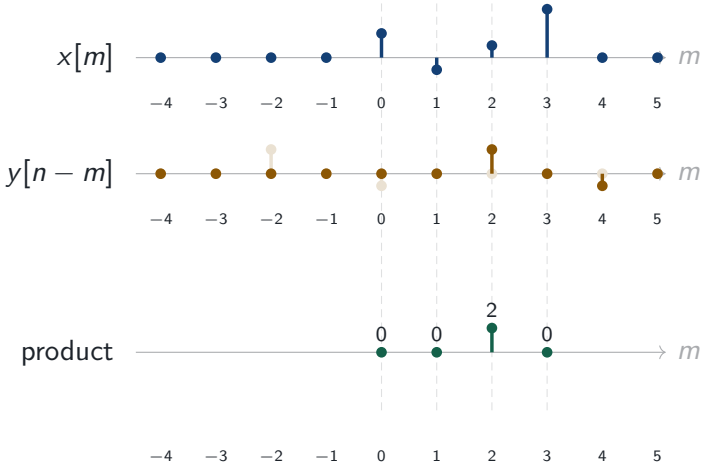
- Keep $x[m]$ fixed and reverse, and shift y



$$n = 3$$
$$-2 - 4 = -6$$

4(a): Computing the Linear Convolution

- Keep $x[m]$ fixed and reverse, and shift y

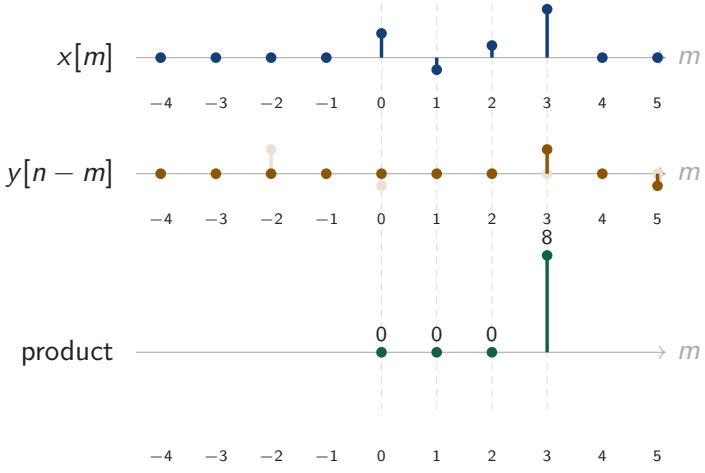


$n = 4$

2

4(a): Computing the Linear Convolution

- Keep $x[m]$ fixed and reverse, and shift y



$n = 5$

8

4(a): Linear Convolution

- We can also use the formula below:

$$(x * y)[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m].$$

- The six non-zero values in the linear convolution are

$$\boxed{x * y = [-2, 1, 3, -6, 2, 8].}$$

- The nonzero output runs from $n = 0$ to $n = 5$

Recall: Circular Convolution

- The circular convolution was defined alternatively:

$$(x \circledast_4 y)[n] = \sum_{m=0}^{3} x[m]y[(n-m) \bmod 4].$$

- First, periodically repeat x and y with period N
- Reverse, and shift y and multiply with x are before
- Add the products over **one period only**

Question 4(b): Inverting Products of DFTs

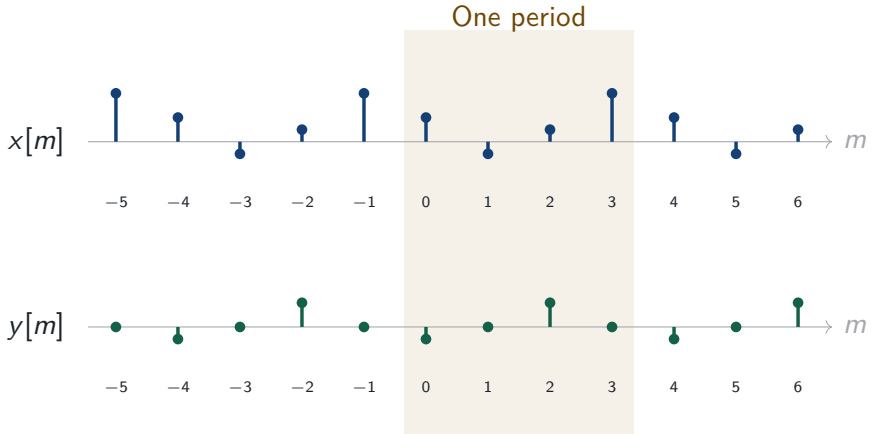
- For the same signals

$$x = [2, -1, 1, 4], \qquad y = [-1, 0, 2],$$

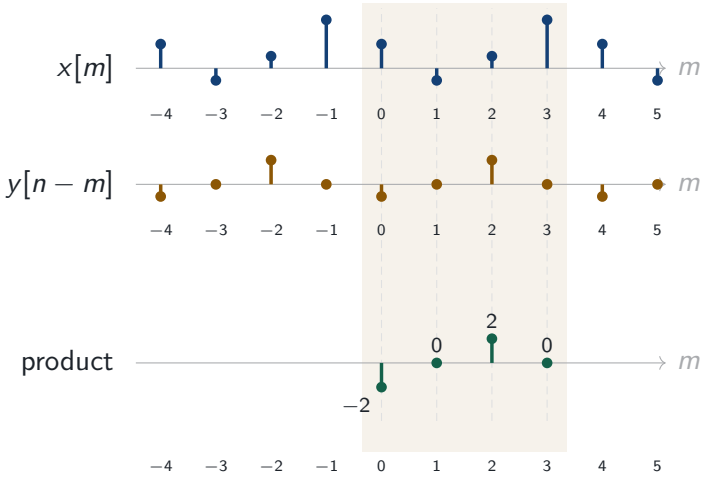
  let X and Y be their DFTs with $N = 4$.
- **Without computing the DFTs**, find the inverse DFT of $X[k]Y[k]$
- DFT products implement **circular convolution**

4(b): Periodic Extensions

- For $N = 4$, repeat $[2, -1, 1, 4]$ and $[-1, 0, 2, 0]$.

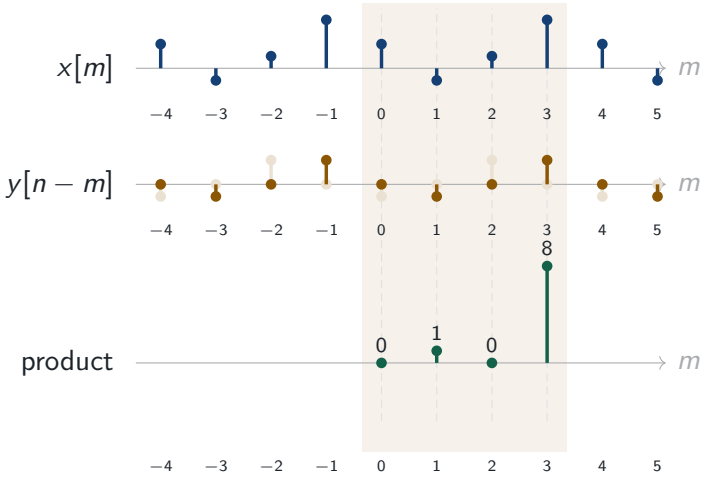


4(b): Computing Circular Convolution



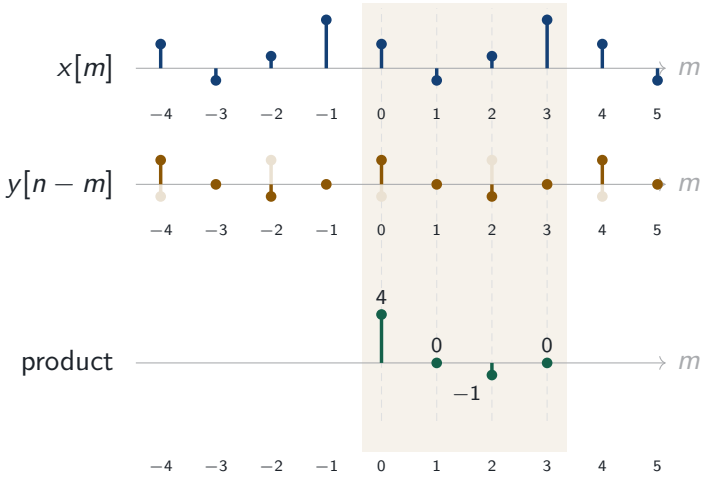
$$n = 0$$
$$-2 + 2 = 0$$

4(b): Computing Circular Convolution



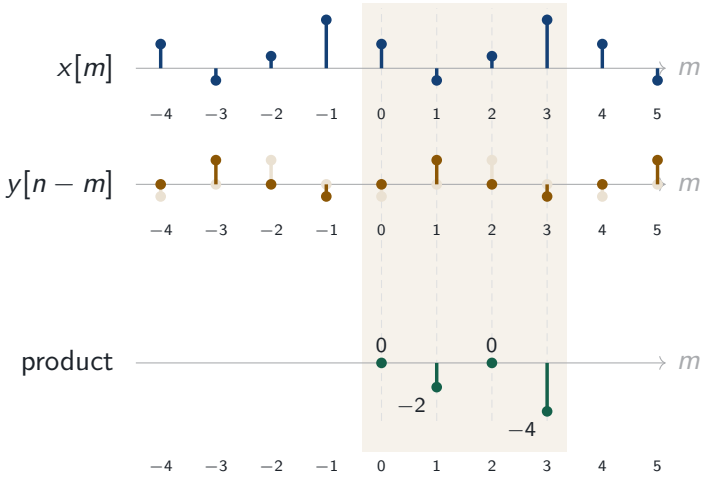
$$n = 1$$
$$1 + 8 = 9$$

4(b): Computing Circular Convolution



$n = 2$
$4 - 1 = 3$

4(b): Computing Circular Convolution



$n = 3$

$-2 - 4 = -6$

4(b): Circular Convolution

- As we see, this operation implements the circular convolution

$$(x \circledast_4 y)[n] = \sum_{m=0}^{3} x[m]y[(n-m) \bmod 4].$$

- Therefore the inverse DFT corresponds to:

$$\boxed{\text{IDFT}_4\{X[k]Y[k]\} = [0, 9, 3, -6].}$$

Question 4(c): Relate the Two Convolutions

- How can you obtain the circular convolution from the linear convolution?
- What is the relationship between the two results?
- Recall the circular and linear convolutions:

$$C = [0, 9, 3, -6] \qquad \text{and} \qquad L = [-2, 1, 3, -6, 2, 8]$$

4(c): Combining the Cicular and Linear Convolutions

- Recall the circular and linear convolutions:

$$C = [0, 9, 3, -6] \qquad \text{and} \qquad L = [-2, 1, 3, -6, 2, 8]$$

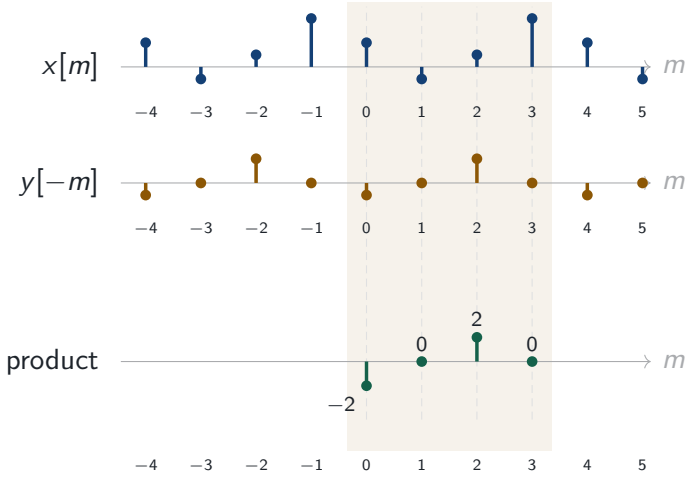
- We see that:

$$C[0] = L[0] + L[4]$$
$$C[1] = L[1] + L[5]$$

- The remaining values are the same in both
- Why does this happen?

4(c): Decomposing a Circular Convolution

- At $n = 0$, add the four products in one period.

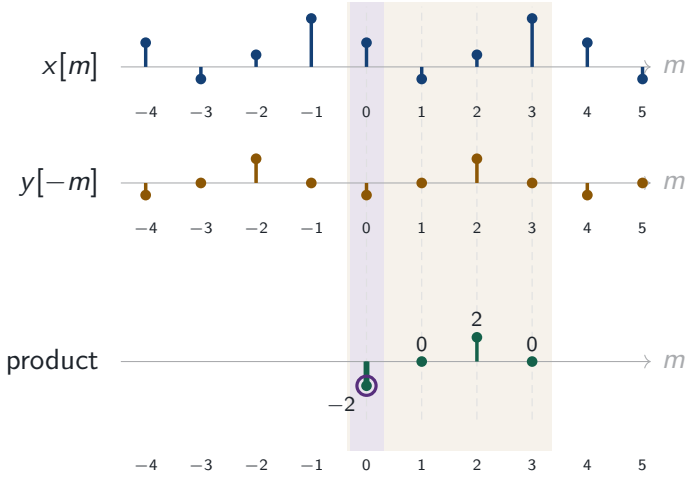


$n = 0$

$-2 + 2 = 0$

4(c): Decomposing a Circular Convolution

- The highlighted product is the one from linear output $n = 0$.

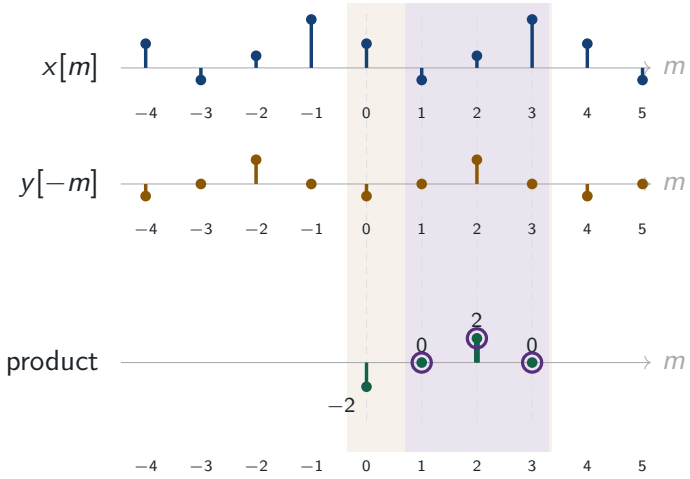


$n = 0$
$-2 + 2 = 0$

Linear: $n = 0$
$2(-1) = -2$

4(c): Decomposing a Circular Convolution

- The three highlighted products are from linear output $n = 4$.



$n = 0$
$-2 + 2 = 0$

Linear: $n = 4$

$(-1)(0) = 0$

$1(2) = 2$

$4(0) = 0$

4(c): Circular Convolution from Linear Convolution

- In general if x and y are supported on $0, \ldots, N-1$
- Then, for $0 \leq n < N$:

$$\boxed{(x \circledast_N y)[n] = (x * y)[n] + (x * y)[n+N].}$$

- This can also be seen from the formula

4(c): Circular Convolution from Linear Convolution

- This can also be seen from the formula:

$$
\begin{aligned}
(x \circledast_N y)[n] &= \sum_{m=0}^{N-1} x[m]y[(n-m) \bmod N] \\
&= \sum_{m=0}^{n} x[m]y[(n-m) \bmod N] + \sum_{m=n+1}^{N-1} x[m]y[(n-m) \bmod N] \\
&= \sum_{m=0}^{n} x[m]y[n-m] + \sum_{m=n+1}^{N-1} x[m]y[N+(n-m)]
\end{aligned}
$$

- Observe that the last terms correspond to two linear convolutions
- The remaining terms in the linear convolution sums are 0

Question 4(d): Zero-Padding

- Start with $x = [2, -1, 1, 4]$ and $y = [-1, 0, 2]$.
- How many zeros should you append to **each** signal so that circular convolution recovers their linear convolution?
- Justify your answer using the relationship from part (c)

4(d): Padding the Inputs

- We require length $L_x + L_y - 1$
- For us $L_x = 4, L_y = 3$
- Hence, we need to pad to length 6
- Append **two zeros to** x and **three zeros to** y:

$$\widetilde{x} = [2, -1, 1, 4, 0, 0], \qquad \widetilde{y} = [-1, 0, 2, 0, 0, 0].$$

4(d): Why Does Padding Work?

- For $0 \le n < 6$, we use the relationship from part (c):

$$(\widetilde{x} \circledast_6 \widetilde{y})[n] = (x * y)[n] + \underbrace{(x * y)[n+6]}_{0} = (x * y)[n].$$

- DFT products of the padded inputs therefore give

$$\boxed{\text{IDFT}_6\{\widetilde{X}[k]\widetilde{Y}[k]\} = [-2, 1, 3, -6, 2, 8].}$$